MAT1102: Differential Calculus and Coordinate Geometry

Final-term Examination
SAMPLE QUESTION

1. Given that $f(x) = \begin{cases} x^2, x \leq 2 \\ x+2, x > 2 \end{cases}$,
   (a) sketch $f(x)$,
   (b) find $\lim_{x\to 2} f(x)$,
   (c) discuss the continuity of $f(x)$ at $x = 2$.

2. Given that $f(x) = \begin{cases} 2-x, & \text{for } x \leq 2 \\ x-2, & \text{for } x > 2 \end{cases}$ at $x = 2$.
   (a) sketch $f(x)$,
   (b) discuss the continuity of $f(x)$ at $x = 2$,

3. Find $\frac{dy}{dx}$ of the followings:
   (a) $y = \tan x - 5x^4 + e^{2x}$, (b) $y = x^{\sin 2x}$, (c) $x = t - \cos t,\ y = \sin t$,
   (d) $x^2y^2 + 5x^2y - 3x = 7$, (e) $y = \frac{x^2+1}{x+2}$.

4. Given that $f(x) = x^3 + 2x + 3$,
   (a) find $f'(x)$ and $f''(x)$,
   (b) find $f''(2)$,
   (c) find the equations of tangent and normal line to the graph of $f(x)$ at $x = -1$.

5. Find the equations of tangent and normal line to the graph of $x^4 + 2y + x^2y = x + 2$ at the point $(0,1)$.

8. Find the equations of tangent and normal lines to the curve $y = \sqrt{x}$ at the point $(1,1)$.

9. Find the first three non-zero terms of the Taylor series expansion of $f(x) = \cos 2x$ about $x = \frac{\pi}{4}$.

10. Evaluate of the following limits (if possible):

(a) $\lim_{x\to 0^+} x\ln x$, (b) $\lim_{x\to 0}\left(\frac{1}{x^2}-\frac{\cos 3x}{x^2}\right)$, (c) $\lim_{x\to 0^+}\left(e^{3x}-x\right)^{\frac{1}{x}}$.

11. Given that $f(x)=x^3-9x^2+15x+1$,

(i) the stationary points and the points of inflection,

(ii) the intervals in which $f(x)$ is increasing and decreasing,

(iii) discuss the nature of stationary points and,

(iv) sketch the curve using all the above information.

12. Find all the relative maxima, relative minima, and saddle points (if any) of the following functions:

(a) $z=x^2y^2-x^2-y^2$, (b) $z=2x^3-6xy+3y^2$,